

Ex:No: 17

HIGH LEVEL PROGRAMMING EXTENSIONS-FUNCTIONS

1)

```
mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE FUNCTION GetCustomerLevel(p_creditLimit DECIMAL(10,2))
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
->     DECLARE customer_level VARCHAR(20);
->
->     IF p_creditLimit > 50000 THEN
->         SET customer_level = 'PLATINUM';
->     ELSEIF p_creditLimit >= 10000 AND p_creditLimit <= 50000 THEN
->         SET customer_level = 'GOLD';
->     ELSE
->         SET customer_level = 'SILVER';
->     END IF;
->
->     RETURN customer_level;
-> END //
```

Query OK, 0 rows affected (0.226 sec)

```
mysql>
mysql> DELIMITER ;
mysql>
mysql> SELECT customerNumber, customerName, creditLimit,
->         GetCustomerLevel(creditLimit) AS CustomerLevel
-> FROM customers;
```

customerNumber	customerName	creditLimit	CustomerLevel
101	Arun	75000.00	PLATINUM
102	Priya	30000.00	GOLD
103	Rahul	5000.00	SILVER

3 rows in set (0.191 sec)

2)

```

mysql>
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE Factorial(
    ->     IN n INT,
    ->     OUT result BIGINT
    -> )
    -> BEGIN
    ->     IF n <= 1 THEN
    ->         SET result = 1;
    ->     ELSE
    ->         CALL Factorial(n - 1, @temp);
    ->         SET result = n * @temp;
    ->     END IF;
    -> END //

```

Query OK, 0 rows affected (0.354 sec)

```

mysql>
mysql> DELIMITER ;
mysql>
mysql> CALL Factorial(5, @result);
Query OK, 0 rows affected (0.122 sec)

```

```

mysql> SELECT @result AS Factorial;

```

```

+-----+
| Factorial |
+-----+
|         120 |
+-----+

```

1 row in set (0.092 sec)